

Review retrospective:
*the integral-remainder foundation — a clean pass on the
primer’s covariance formula, and a 2668-line file that had
outgrown its own header*

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

2026-06-05

Abstract

A soundness review of `Laplace/OneD/IntegralRemainder.lean` — at 2668 lines the largest file in the laplace seabed, and the analytic foundation on which the whole 1D anharmonic programme rests: the I_n/J_n integral-remainder asymptotics culminating in the SLT primer’s Stage-2 covariance formula $\text{Cov}_t[w^2, w] = -2\alpha/(\lambda^3 t^2) + o(t^{-2})$ (eq. 4.10). Result: a **clean fidelity pass on every headline**, with ten landed edits — two dead opens, four docstring corrections, and four dead locals — and several larger refactors deliberately left as forward tides. Reviewing it completes independent coverage of the laplace `OneD` directory.

1 Setting

The capstone of a session-long sweep of the laplace `OneD` directory. Almost every other `OneD` file — the anharmonic cumulant arc, the harmonic stack, the bounded-prior pair — ultimately consumes the asymptotics proved here. The file is the bridge from the elementary Gaussian moments of Stage 1 to the first genuine payoff of Stage 2.

2 What was reviewed

The headline chain: a pointwise Taylor-remainder bound for $e^{-s} - (1 - s)$ against the Gaussian weight; polynomial $\times$ Gaussian integrability; the linearised integrand decomposition; the J_n first-order expansions $J_n = M_n - (A/\sqrt{t})M_{n+3} - (B/t)M_{n+4} + O(1/t)$ with Gaussian moments $M_{2k} = (2k - 1)!!\sqrt{2\pi}$; the $I_n \leftrightarrow J_n$ substitution identity $\sqrt{\lambda}t^{n+1}I_n = J_n$; the I_n asymptotics; `cov_coefficient_identity`; and the assembled `cov_anharmonic_asymptotic`, `mean_anharmonic_asymptotic`, `var_anharmonic_J_form_asymptotic`, `cov_self_anharmonic_asymptotic`, and the explicit $O(t^{-2})$ rate theorems. Ground truth: the primer’s eq. 4.10 and the file’s own docstrings, plus internal consistency of the moment/substitution algebra.

3 Fidelity / soundness findings

Sound; no defect in any headline. The reviewer verified, item by item: the covariance coefficient $-2\alpha/\lambda^3$ (sign and power t^2 correct); the mean $-\alpha/(2\lambda^2)$; the variance and self-covariance

$1/\lambda$; both explicit $O(t^{-2})$ rate statements; the Gaussian normalisations $\sqrt{2\pi}$, $3\sqrt{2\pi}$, $15\sqrt{2\pi}$ and the vanishing odd moments; the J_1, J_3 constants $-3A\sqrt{2\pi}$, $-15A\sqrt{2\pi}$; and the substitution identity’s Jacobian/power. “No defect found” anywhere in the constants, signs, exponents, or the Jacobian.

One honest under-statement (report-only). The specialised J_0/J_2 and I_0/I_2 theorems state only the leading term plus $O(1/t)$, absorbing the explicit quartic first-correction coefficient that J_n.asymptotic could expose. These are true as written; *strengthening* them to surface the coefficient is a stronger statement — an API change, hence a forward tide, not a review edit.

4 Simplifications and documentation fixes applied

Two dead opens. `open Real` and `open Set` both carried nothing — every `Real.*` and every interval `(Set.Ioi/Set.Iic)` is qualified; the bare `exp/√/π` tokens live only in comments. `open MeasureTheory` (166 bare `∫`) and `open scoped Nat` (the `!!`) stay.

Four docstring corrections — the file had outgrown its own header. The module docstring still described only the opening pointwise bound and asserted “no new analytic content beyond the algebra,” for a file that in fact proves the entire covariance result; it was retitled and the false claim corrected. A section header “Specialised asymptotics for $n = 1$ ” actually covers J_0 – J_3 . A summary block claimed the covariance assembly was “left as a follow-up” — it is proved later in the very same file. And the `cov_anharmonic_J_form_asymptotic` docstring self-contradicted (“multiply by $\lambda^{3/2}$ ” then “dividing by $\lambda^{3/2}$ ”); the proof divides (`div_const`), so the prose was fixed to “divides” and clarified that the theorem states the J_n -quotient limit.

Four dead locals. Three unused `haves` in `J_0_asymptotic` (`habs`, `hpos_bound`, `h_quartic_abs` — each defined then never referenced; the `calc` that follows uses none) and one unused `with hM_def` equation name on a `set`. All ten edits build green (8254 jobs), no new in-file warnings, exported signatures byte-identical; the reviewer affirmed each.

5 Disagreements and open questions

None substantive. The `import Laplace.OneD.Anharmonic` is transitively redundant (its `anharmonicPotential` is re-exported via `Rescaling`, which imports `Anharmonic` directly), but it was *kept*: `anharmonicPotential` is used directly and pervasively here, so the import is a heavily-used *direct* dependency that merely happens to be reachable two ways — removing it would make a real dependency implicit (a clarity-reducing refactor), not delete dead code. The reviewer agreed it was “not clearly dead.”

6 Lessons

- **The largest files drift the most in their headers, and the module docstring drifts first.** A file that grows section-by-section over many tides keeps the docstring it was born with: here a 2668-line covariance proof still announced itself as a pointwise-bound lemma with “no new analytic content.” On a big mature file, budget the fidelity check to (i) the handful of headline *statements* and (ii) the *navigational* docstrings (module header, section headers, summary blocks) — those, not the routine integrability lemmas, are where the drift hides.
- **“Redundant import” $\neq$ “dead import” when the module’s symbols are used directly.** The dead-import test is “is any symbol of this module used here?”, not “is this module

reachable another way?”. `Anharmonic` fails the second but passes the first emphatically — keep it. (Contrast a genuinely symbol-less import, which the build lets you drop.)

- **A self-contradicting docstring is an unambiguous fix.** “Multiply by $\lambda^{3/2}$ ” and “divide by $\lambda^{3/2}$ ” cannot both be right; the proof term (`div_const`) breaks the tie. Internal contradiction is the easiest docstring defect to adjudicate — no external ground truth needed.

7 Follow-ups

- Forward tide: strengthen `J_0/J_2/I_0/I_2` to expose the explicit quartic first-correction coefficient (currently absorbed into $O(1/t)$).
- Batched-hygiene tide: the pre-existing style-linter warnings (`show-for-readability` $\times 5$, flexible `simp` $\times 2$, one over-long line), the remaining unused `with hX.def` equation names, and the proof-dedup (`mean/cov_self` redo the `*_J_form_exact` algebra; the repeated $\sqrt{}$ -substitution block).
- With this review, every substantive file in laplace `OneD` has been independently reviewed.